

HW 06 SOLUTIONS

Practice Problems

3.67

$$(0.7)^4(0.3) = 0.07203$$

3.71

(a)

$$P(Y > a) = \sum_{y=a+1}^{\infty} q^{y-1}p = pq^a \sum_{y=1}^{\infty} q^{y-1} = \frac{pq^a}{1-q} = q^a$$

(b) From part (a),

$$P(Y > a+b \mid Y > a) = \frac{P(Y > a+b)}{P(Y > a)} = \frac{q^{a+b}}{q^a} = q^b$$

(c)

$$P(Y > a+b \mid Y > a) = P(Y > b)$$

(d) The results in the past are not relevant to a future outcome (independent trials).

3.73

Let Y = number of accounts audited until the first with substantial errors is found.

(a)

$$P(Y = 3) = (0.12)(0.9)^2 = 0.009$$

(b)

$$P(Y \geq 3) = P(Y > 2) = (0.9)^2 = 0.81$$

3.77

$$\begin{aligned}
P(Y = \text{odd}) &= P(Y = 1, 3, 5, \dots) = P(Y = 2k + 1 \text{ for integers } k = 1, 2, \dots) = \sum_{k=1}^{\infty} q^{2k+1-1}p \\
&= p \sum_{k=1}^{\infty} q^{2k} = p \sum_{k=1}^{\infty} (q^2)^k \\
&= \frac{p}{(1 - q^2)}
\end{aligned}$$

3.93

From Ex. 3.92:

(a)

$$P(Y = 5) = \binom{4}{2} (0.9^3)(0.1^2) = 0.04374$$

(b)

$$P(Y \leq 5) = P(Y = 3) + P(Y = 4) + P(Y = 5) = 0.729 + 0.2187 + 0.04374 = 0.99144$$

3.97

(a) Geometric probability:

$$(0.8)^2(0.2) = 0.128$$

(b) Negative binomial probability:

$$\binom{6}{2} (0.2^3)(0.8^4) = 0.049$$

(c) The trials are independent and the probability of success is the same from trial to trial.

(d)

$$\mu = \frac{3}{0.2} = 15, \quad \sigma^2 = \frac{3(0.8)}{0.2^2} = 60$$

3.103

Use the hypergeometric distribution with $N = 10, r = 4, n = 5$:

$$P(Y = 0) = \binom{6}{5} \binom{10}{5} = 0.0238$$

3.105

- (a) Y follows a hypergeometric distribution. The probability of being chosen on a trial is dependent on previous outcomes.

(b)

$$P(Y \geq 2) = P(Y = 2) + P(Y = 3) = 0.5357 + 0.1786 = 0.7143$$

(c)

$$\mu = 3 \cdot \frac{5}{8} = 1.875, \quad \sigma^2 = 3 \cdot \frac{5}{8} \cdot \frac{3}{8} \cdot \frac{5}{7} = 0.5022, \quad \sigma = 0.7087$$

3.121

(a)

$$P(Y = 4) = \frac{\lambda^4 e^{-\lambda}}{4!} = 0.090$$

(b)

$$P(Y \geq 4) = 1 - P(Y \leq 3) = 1 - 0.857 = 0.143$$

(c)

$$P(Y < 4) = P(Y \leq 3) = 0.857$$

(d)

$$P(Y \geq 4 \mid Y \geq 2) = \frac{P(Y \geq 4)}{P(Y \geq 2)} = \frac{0.143}{0.594} = 0.241$$

3.123

If $p(0) = p(1)$, then

$$\frac{\lambda^0 e^{-\lambda}}{0!} = \frac{\lambda^1 e^{-\lambda}}{1!}$$

So $\lambda = 1$.

Then

$$p(2) = \frac{1^2 e^{-1}}{2!} = 0.1839$$

Submitted Problems

3.70

Let Y = number of holes drilled until a productive well is found.

(a)

$$P(Y = 3) = (0.8)^2(0.2) = 0.128$$

(b)

$$P(Y > 10) = (0.8)^{10} = 0.107$$

3.80

Let Y = number of tosses until the first 6 appears, so $Y \sim \text{Geometric}(p = 1/6)$.

From Ex. 3.77,

$$P(\text{B tosses first 6}) = P(Y = 2, 4, 6, \dots) = 1 - P(Y = 1, 3, 5, \dots) = 1 - \frac{p}{1 - q^2}$$

With $p = 1/6$, this becomes $5/11$.

Then

$$P(Y = 4 \mid \text{B tosses first 6}) = \frac{(5/6)^3(1/6)}{5/11} = 275/1296$$

3.92

Let Y = number of trials until the first non-defective engine is found. $Y \sim \text{Geometric}(p = 0.9)$.

$$P(Y = 2) = (0.1)(0.9) = 0.09$$

3.94

(a)

$$\mu = \frac{1}{0.9} = 1.11, \quad \sigma^2 = \frac{0.1}{0.9^2} = 0.1234$$

(b)

$$\mu = \frac{3}{0.9} = 3.33, \quad \sigma^2 = \frac{3(0.1)}{0.9^2} = 0.3704$$

3.106

Using results from Ex. 3.103:

$$E(50Y) = 50E(Y) = 50 \cdot 2 = 100$$

$$V(50Y) = 2500V(Y) = 2500 \cdot 0.6667 = 1666.67$$

3.108

Use the fact that $P(\text{at least one is defective}) = 1 - P(\text{none are defective})$. Then, we require $P(\text{none are defective}) \leq 0.2$. If $n = 8$,

$$P(\text{none are defective}) = \left(\frac{17}{20}\right) \left(\frac{16}{19}\right) \left(\frac{15}{18}\right) \left(\frac{14}{17}\right) \left(\frac{13}{16}\right) \left(\frac{12}{15}\right) \left(\frac{11}{14}\right) \left(\frac{10}{13}\right) = 0.193.$$

3.120

There are N animals in this population. After taking a sample of k animals, marking and releasing them, there are $N - k$ unmarked animals. We then choose a second sample of size 3 from the N animals. There are $\binom{N}{3}$ ways of choosing this second sample and there are $\binom{N-k}{2}\binom{k}{1}$ ways of finding exactly one of the originally marked animals. For $k = 4$, the probability of finding just one marked animal is

$$P(Y = 1) = \frac{\binom{N-4}{2}\binom{4}{1}}{\binom{N}{3}} = \frac{12(N-4)(N-5)}{N(N-1)(N-2)}.$$

Calculating this for various values of N , we find that the probability is largest for $N = 11$ or $N = 12$ (the same probability is found: 0.503).

3.128

Over one minute, $Y \sim \text{Poisson}(\lambda = 80/60 = 4/3)$.

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - e^{-4/3} = 0.7364$$

3.138

$$\begin{aligned} E(Y(Y-1)) &= \sum_{y=0}^{\infty} \frac{y(y-1)\lambda^y e^{-\lambda}}{y!} \\ &= \sum_{y=2}^{\infty} \frac{y(y-1)\lambda^y e^{-\lambda}}{y!} \\ &= e^{-\lambda} \sum_{y=2}^{\infty} \frac{\lambda^y}{(y-2)!} \frac{\lambda^2}{\lambda^2} \\ &= e^{-\lambda} \lambda^2 \sum_{y-2=0}^{\infty} \frac{\lambda^{y-2}}{(y-2)!} \\ &= e^{-\lambda} \lambda^2 e^{\lambda} \\ &= \lambda^2 \end{aligned}$$

Then

$$E[Y(Y-1)] = E(Y^2) - E(Y) = \lambda^2 \implies E(Y^2) = \lambda^2 + \lambda$$

$$\text{And } V(X) = E(Y^2) - [E(Y)]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$